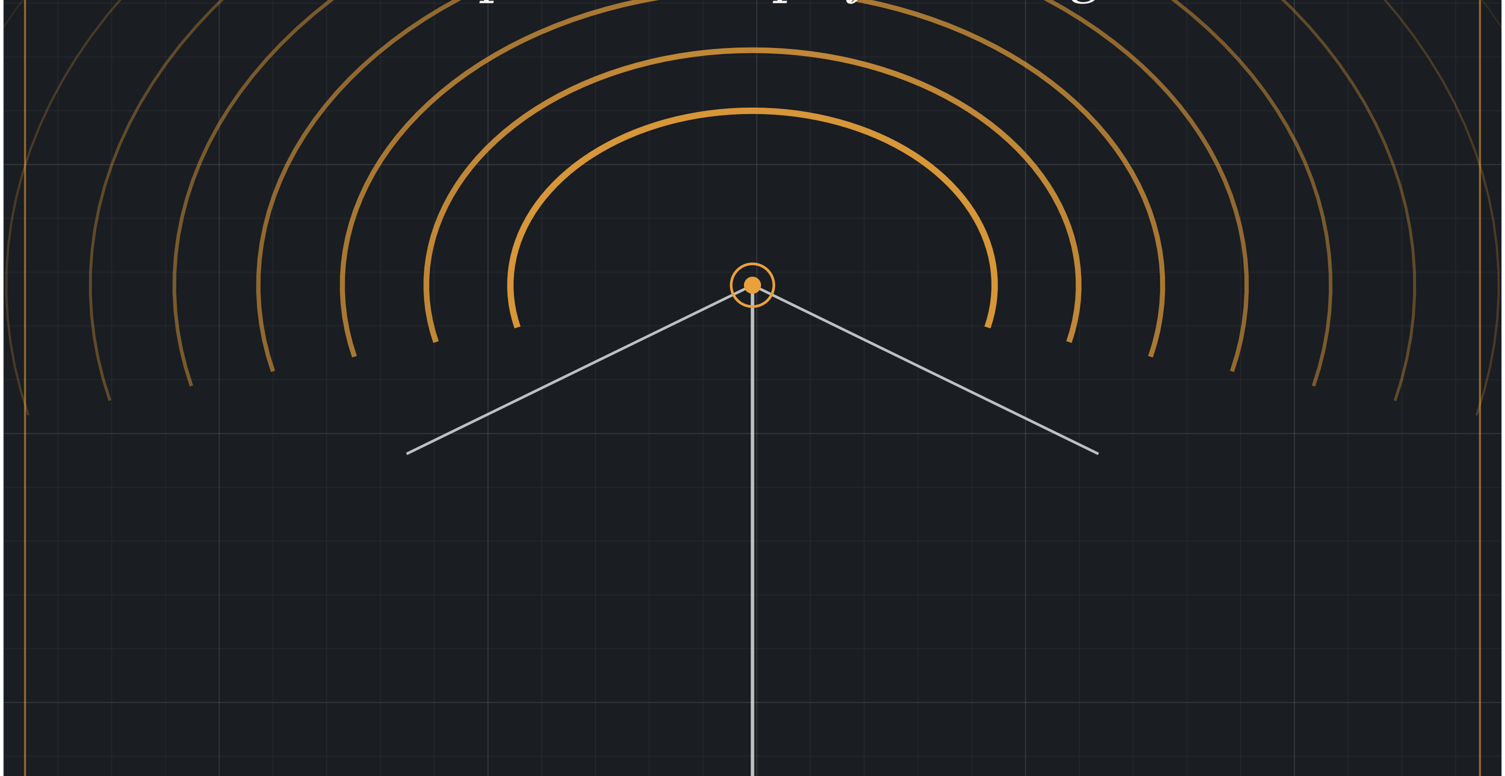
PORTABLE OPERATION · SITE, SIGNAL, DEBRIEF

FIELD RECORD

The Portable Operator's Deployment Log



SAMPLE "NOCALL" STATION

26 DEPLOYMENTS · FIELD REFERENCE · STATION RECORDS

FIELD RECORD

A logbook built around the outing, not the contact.

SAMPLE "N0CALL" STATION

FIELD RECORD

The Portable Operator's Deployment Log

26 DEPLOYMENTS · SITE, SIGNAL, AND DEBRIEF
FIELD REFERENCE FOR PORTABLE OPERATION

SAMPLE "NOCALL" STATION

FIELD RECORD: The Portable Operator's Deployment Log

Compiled and designed by Sample "N0CALL" Station.

Frequency allocations, award-program requirements, and regulatory notes in Part One were compiled from the sources listed in the Colophon. Rules change. The operator, not the book, is responsible for lawful operation. Before you transmit, check 47 CFR Part 97 and the current rules of any award program you are chasing.

Nothing in this book is legal advice, engineering certification, or a substitute for your own RF exposure evaluation.

FACTS AND SOURCES
Frequency allocations, repeater listings, tone codes and program rules are facts and belong to nobody. The selection, arrangement, drawings and wording here are original to this work. Sources are named on the page that uses them and in the Colophon.

BUILD
This copy was generated on 08 September 2026. Regulatory figures were current when the sources in the Colophon were compiled. A copy printed long after that date should be checked against the rules before it is trusted.

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CONTENTS

HOW TO USE THIS BOOK

A conventional logbook records the contact and assumes the station. For a fixed station that assumption holds. For a portable one it does not: the antenna, the support, the power source, the noise floor and the weather change with every outing, and they are the variables that determine what the log shows.

A contact with Colorado on 20 meters at 40 watts is not a complete record. The same contact made from a wire at 22 feet, on a battery two hours into its discharge, at a site 60 feet from a power line, with the K index at 4, is. Without the second set of facts the first cannot be interpreted later, and interpretation is the reason to keep a log at all.

The unit of record is the deployment

Part Two is organized into deployments. Each occupies four pages, laid out as two facing spreads, and is completed over the course of a single outing:

- **Site Record**, left page of the first spread. Location, conditions, station, antenna and power. Complete it before the first call, while the details are in front of you.
- **QSO Log A**, right page of the first spread. Contacts, in columns chosen for portable operating.
- **QSO Log B**, left page of the second spread. Continuation. If both sheets fill, continue on the overflow pages in Part Three and note the deployment number.
- **Debrief**, right page of the second spread. Totals by band and mode, power consumed, faults, and the change to make before the next outing. Complete it at the site.

Scope of the written record

Record what cannot be reconstructed afterwards: coordinates, feedpoint height, wire configuration, measured SWR, noise floor, and the point at which the supply voltage fell. Where an electronic log is in use it holds the contact list; these pages hold the context around it and serve as the running record during operation.

Times are UTC

Every date and time field in this book is Coordinated Universal Time. Award programs, contest logs and net schedules are all keyed to it, and the UTC day boundary falls during the evening in North America. A conversion table is given in Part One.

Part One, reference

Band edges, calling frequencies, wire lengths, feedline loss, power budgets, exposure limits and activation requirements. Read once, then consulted. Each section carries a marker at a distinct height in the outer margin, so the sections can be located from the fore edge.

Part Three, station records

Equipment register, modification and repair log, antenna build sheets with SWR sweep grids, battery cycle records and award trackers. These carry forward from one volume to the next.

> Number every deployment, including unsuccessful ones. An outing that produced three contacts because a feedpoint connector failed carries more diagnostic value than one that produced a hundred without incident, and only the debrief preserves it.

OPERATOR

CALLSIGN | NAME | LICENSE CLASS | EXPIRES

HOME GRID | ARRL SECTION | FRN | CLUB

EMAIL | QSL ROUTE | LOTW | LOGBOOK SOFTWARE

PROGRAMS AND IDENTIFIERS

POTA ACCOUNT | SOTA ACCOUNT | WWFF | CONTEST SECTION

IF FOUND

This is a working logbook. It has no commercial value and considerable value to its owner. The operator would be glad to hear from the finder.

CONTACT | PHONE | REWARD

EMERGENCY INFORMATION

EMERGENCY CONTACT | RELATIONSHIP | PHONE

MEDICAL NOTES FOR FIRST RESPONDERS

VEHICLE | PLATE | LOCAL REPEATER / NET

GO-KIT MANIFEST

EQUIPMENT CARRIED

Enter the standing kit once and check it in pencil, or reproduce this page for each outing.

ITEM	DETAIL / MODEL	QTY	BAG	CHK

PART ONE

FIELD REFERENCE

The material a portable operator reaches for with cold hands. Read it once at the kitchen table. After that, use the markers in the outer margin.

US BAND PLAN — GENERAL PRIVILEGES

SOLID BARS ARE YOURS TO TRANSMIT ON AS A GENERAL. OUTLINED BARS BELONG TO A HIGHER CLASS. EACH BAND IS ON ITS OWN SCALE, FREQUENCIES IN MHZ.

YOURS TO USE　HIGHER CLASS ONLY

1500 W PEP unless marked. 30 m is 200 W and carries no phone. 60 m is secondary: 9.15 W ERP referenced to a dipole, 2.8 kHz bandwidth, General and above, and the four legacy channels either side of the subband survive at 100 W ERP. Technician and Novice are 200 W PEP on the HF slices shown, and 25 W PEP on 1.25 m. Compiled from 47 CFR Part 97 and ARRL allocation data, current 2026. Your class carries its own power limits; the outlined segments are not yours until you upgrade. Verify before you transmit.

01 BANDS

US Band Plan and Privileges

Sub-band edges are not marked on the dial. In the field, with cold hands and a knob that is easy to bump, they have to be known.

Sub-band edges below are US allocations under 47 CFR 97.301 and 97.305, current as of 2026. E = Amateur Extra, A = Advanced, G = General, T = Technician, N = Novice. Unless a lower limit is shown, the maximum is 1500 W PEP output, and the standing rule always applies: use the minimum power necessary.

Low frequency and 160 meters

BAND	FREQUENCY	MODES	CLASS	NOTES
2200 m	135.7 - 137.8 kHz	CW, data	T A G E	1 W EIRP. Advance notice to UTC required.
630 m	472 - 479 kHz	CW, data	T A G E	5 W EIRP. Advance notice to UTC required.
160 m	1.800 - 2.000 MHz	CW, phone, image, data	G A E	No sub-band split. Regional band plan applies.

80 and 75 meters

BAND	FREQUENCY (MHZ)	MODES	CLASS	NOTES
80 m	3.500 - 3.525	CW, data	E	
	3.525 - 3.600	CW, data	T N A G E	T and N: CW only, 200 W PEP.
	3.600 - 3.700	CW, phone, image, data	E	
75 m	3.700 - 3.800	CW, phone, image	A E	
	3.800 - 4.000	CW, phone, image	G A E	

60 meters, as revised 13 February 2026

The 60 meter band changed shape in 2026. A contiguous 15 kHz subband replaced channelized operation in the middle of the old channel group, and the former 5358.5 kHz channel disappeared because it now falls inside that subband and would otherwise carry two different power limits. Four legacy channels survive on either side. Everything on 60 meters is secondary: you accept interference from the primary government users and you must not cause any.

BAND	FREQUENCY	MODES	CLASS	NOTES
60 m	5351.5 - 5366.5 kHz	Any mode, 2.8 kHz max BW	G A E	9.15 W ERP max. Secondary.
	5332.0 kHz ctr (5330.5 USB)	USB, CW, data	G A E	100 W ERP. Channel.
	5348.0 kHz ctr (5346.5 USB)	USB, CW, data	G A E	100 W ERP. Channel.
	5373.0 kHz ctr (5371.5 USB)	USB, CW, data	G A E	100 W ERP. Channel.
	5405.0 kHz ctr (5403.5 USB)	USB, CW, data	G A E	100 W ERP. Channel.

> ERP on 60 meters is referenced to a half-wave dipole, not to an isotropic radiator and not to your transmitter output. If your antenna has gain relative to a dipole, you must reduce transmitter power to stay inside the limit. A compromised portable wire usually has loss, not gain, but you are responsible for knowing which.

BANDS

40 and 30 meters

BAND	FREQUENCY (MHZ)	MODES	CLASS	NOTES
40 m	7.000 - 7.025	CW, data	E	
	7.025 - 7.125	CW, data	T N A G E	T and N: CW only, 200 W PEP.
	7.125 - 7.175	CW, phone, image	A E	
	7.175 - 7.300	CW, phone, image	G A E	
30 m	10.100 - 10.150	CW, data only	G A E	200 W PEP max. No phone, no image.

20 and 17 meters

BAND	FREQUENCY (MHZ)	MODES	CLASS	NOTES
20 m	14.000 - 14.025	CW	E	
	14.025 - 14.150	CW, data	G A E	
	14.150 - 14.175	Phone, image	E	
	14.175 - 14.225	Phone, image	A E	
	14.225 - 14.350	Phone, image	G A E	
17 m	18.068 - 18.110	CW, data	G A E	
	18.110 - 18.168	CW, phone, image	G A E	

15, 12, and 10 meters

BAND	FREQUENCY (MHZ)	MODES	CLASS	NOTES
15 m	21.000 - 21.025	CW	E	
	21.025 - 21.200	CW, data	T N A G E	T and N: CW only, 200 W PEP.
	21.200 - 21.225	Phone, image	E	
	21.225 - 21.275	Phone, image	A E	
	21.275 - 21.450	Phone, image	G A E	
12 m	24.890 - 24.930	CW, data	G A E	
	24.930 - 24.990	CW, phone, image	G A E	
10 m	28.000 - 28.300	CW, data	T N A G E	T and N: 200 W PEP.
	28.300 - 28.500	CW, phone	T N A G E	T and N: 200 W PEP.
	28.500 - 29.700	CW, phone, image	G A E	29.5 - 29.7 FM repeaters and simplex.

VHF and above

BAND	FREQUENCY	MODES	CLASS	NOTES
6 m	50.0 - 50.1 MHz	CW only	T A G E	50.090 CW calling. 50.125 SSB calling.
	50.1 - 54.0 MHz	All authorized modes	T A G E	
2 m	144.0 - 144.1 MHz	CW only	T A G E	144.200 SSB calling. 146.520 FM national simplex.
	144.1 - 148.0 MHz	All authorized modes	T A G E	
1.25 m	219 - 220 MHz	Fixed digital point to point	T A G E	Notification required. Not for general use.
	222 - 225 MHz	All authorized modes	T N A G E	N: 222.0 - 225.0, 25 W PEP. 223.500 FM calling.
70 cm	420 - 450 MHz	All authorized modes	T A G E	446.000 FM national simplex. Regional coordination applies.
33 cm	902 - 928 MHz	All authorized modes	T A G E	Secondary. Shared with Part 15 and ISM.
23 cm	1240 - 1300 MHz	All authorized modes	T A G E	Secondary. Coordinate. Segment under study.

Technician HF privileges at a glance

A Technician licensee has full privileges on every amateur band above 50 MHz and five slices of HF. Those slices are enough to do serious portable work, particularly on 10 meters near a solar peak and on 40 meter CW after dark.

BAND	SEGMENT	MODES PERMITTED	POWER
80 m	3.525 - 3.600 MHz	CW only	200 W PEP
40 m	7.025 - 7.125 MHz	CW only	200 W PEP
15 m	21.025 - 21.200 MHz	CW only	200 W PEP
10 m	28.000 - 28.300 MHz	CW and data	200 W PEP
10 m	28.300 - 28.500 MHz	CW and SSB phone	200 W PEP

Emission bandwidth and mode selection

MODE	TYPICAL BANDWIDTH	FIELD NOTES
CW	~100 - 500 Hz	Narrowest practical mode, and the most efficient use of a limited supply.
SSB phone	~2.4 - 2.8 kHz	HF emission bandwidth capped at 2.8 kHz below 28 MHz.
AM	~6 kHz	Permitted where phone is permitted. The carrier consumes most of the output.
FM phone	~16 kHz	29.5 MHz and above. Not permitted in the HF phone sub-bands below 29.0 MHz.
RTTY and data	2.8 kHz max on HF	Symbol rate limits were removed in 2024 and replaced by this bandwidth cap.
Image, SSTV	~3 kHz	Occupies a phone allocation. Announce before transmitting.

02 FREQS

Calling Frequencies and Watering Holes

None of these frequencies are reserved in regulation. They are conventions, and they work because operators expect to find each other on them.

Activity frequencies by band, MHz

ACTIVITY	160	80/75	40	30	20	17	15	12	10
QRP CW	1.810	3.560	7.030	10.106	14.060	18.096	21.060	24.906	28.060
QRP SSB	-	3.985	7.285	-	14.285	18.130	21.385	24.950	28.385
POTA CW	-	3.550	7.032	10.110	14.032	18.082	21.032	24.892	28.032
POTA SSB	-	3.885	7.185	-	14.285	18.155	21.285	24.955	28.420
SOTA CW	-	3.557	7.032	10.110	14.062	18.092	21.062	24.902	28.062
SOTA SSB	-	3.985	7.185	-	14.285	18.155	21.285	24.955	28.420

Listen long enough to establish that the frequency is clear. A weak station below your noise floor is still occupying it. Then ask: "Is this frequency in use?" on phone, or QRL? on CW, twice, before the first call.

Calling channels and standing nets

PURPOSE	FREQUENCY	NOTES
2 m FM national simplex	146.520 MHz	Calling and travel. Move off after contact.
70 cm FM national simplex	446.000 MHz	Calling. Same courtesy applies.
6 m SSB calling	50.125 MHz	Openings are short. Check it often in June.
2 m SSB calling	144.200 MHz	Horizontal polarization. Point the beam.
Maritime Mobile Service Net	14.300 MHz	Also a recognized informal emergency watch frequency.
Hurricane Watch Net	14.325 / 7.268 MHz	Activates only when a storm warrants it. Listen first.
SATERN	14.265 MHz	Salvation Army emergency net.
Winlink and digital gateways	Varies by RMS	Check the current gateway list before you rely on one.

Selecting a run frequency

- Work above the calling frequency rather than on it. Calling frequencies are for establishing contact, not for running.
- On CW, choose a clear 200 Hz rather than a familiar part of the band.
- Announce QRP operation in the exchange. It sets the receiving operator's expectations.
- Self-spot where the program supports it, then hold the announced frequency. Drifting off it costs contacts.

03 OPS

Signal Reports, Q Signals, and Prosigns

The shared vocabulary of an on-air contact: signal reports, the Q signals in common use, and the prosigns sent as single characters on CW.

The RST system

Three digits: readability, strength, tone. Tone is sent only on CW and data. A phone report is two digits. The scale is old, imprecise, and universally understood, which is exactly what a signal report needs to be.

R, READABILITY	S, STRENGTH	T, TONE (CW ONLY)
1 Unreadable	1 Faint, barely perceptible	1 Extremely rough hissing note
2 Barely readable, occasional words	2 Very weak	2 Very rough AC note
3 Readable with difficulty	3 Weak	3 Rough low pitched AC note
4 Readable with practically no difficulty	4 Fair	4 Rather rough AC note
5 Perfectly readable	5 Fairly good	5 Musical, modulated note
	6 Good	6 Modulated note, slight trace of whistle
	7 Moderately strong	7 Near DC note, smooth ripple
	8 Strong	8 Good DC note, trace of ripple
	9 Extremely strong	9 Purest DC note

> S9 on HF is defined as 50 microvolts at a 50 ohm receiver input, with one S unit nominally 6 dB. Few receivers calibrate to it. Above S9 the scale reads in decibels above that reference. Report what the meter shows and note any known optimism of the receiver in the Debrief.

Q signals in daily use

SIGNAL	MEANING	FIELD NOTES
QRA	What is your station name?	Uncommon on the air, common in print.
QRG	What is my exact frequency?	Useful when you suspect drift.
QRL	Is this frequency in use?	Send it twice before you claim a slot.
QRM	Interference from other stations	Man made. Someone else is on top of you.
QRN	Atmospheric noise, static	Natural. Storms, power lines, a bad site.
QRO	Increase power	Costs amp-hours. Consider band or antenna first.
QRP	Reduce power / low power operation	5 W or less output on CW and data, 10 W PEP on SSB.
QRQ	Send faster	Sent by the receiving station, not the sender.
QRS	Send slower	Always honored by an operator worth working.
QRT	Stop sending / closing station	The polite way to leave.
QRU	Do you have anything for me?	Traffic handling.
QRV	I am ready	Also: are you ready?
QRX	Stand by, wait	Give a time if you can. QRX 5 means five minutes.
QRZ	Who is calling me?	Not a general call for contacts, though it is used that way.
QSA	Signal strength, scale 1 to 5	Older than the RST system.
QSB	Fading	The condition that most often explains a lost contact.
QSK	Full break-in, I can hear between elements	Ask before you assume the other station has it.
QSL	Acknowledge receipt, or a confirmation card	Both senses are in daily use.
QSO	A contact	Used as both noun and verb.
QSP	Relay to another station	Traffic handling.
QSY	Change frequency	Give the new frequency. QSY 7.185.
QTC	I have a message to send	Traffic handling.
QTH	Location	Give something a hunter can log: state, park, grid.
QTR	Correct time	UTC unless you say otherwise.

Prosigns

Sent as a single character with no inter-letter space. The overbar in print marks the run-together pair.

PROSIGN	CODE	MEANING
AR	.-.-.	End of transmission or message
AS	.-...	Wait, stand by briefly
BK	-... -.-	Break, invite the other station in
BT	-...-	Separator between paragraphs, written as a dash
CL	-.-. .-..	Closing station, going off the air
KN	-.--.	Go ahead, named station only
K	-.-	Go ahead, anyone
SK	...-.-	End of contact
SN	...-.	Understood
HH	....	Error, disregard and start the word again

CW abbreviations in common use

ABBR	MEANING	ABBR	MEANING
5NN	599, sent fast	GM/GA/GE	Good morning / afternoon / evening
ABT	About	HR	Here
AGN	Again	HW?	How do you copy?
ANT	Antenna	NR	Number, or near
BK	Break	OM / YL	Old man / young lady
CQ	Calling any station	OP	Operator's name
CUL	See you later	PSE	Please
DE	From, this is	PWR	Power
DX	Distance, foreign station	R	Roger, received
ES	And	RIG	Station equipment
FB	Fine business, excellent	RST	Signal report
GD	Good	TU	Thank you
GL	Good luck	UR	Your, you are
GND	Ground	WX	Weather
73	Best regards	88	Love and kisses

04 PHON

Phonetics and Working the Exchange

Standard phonetics exist so that a marginal signal resolves into the correct callsign. Non-standard substitutions are workable on a strong signal and unhelpful on a weak one.

PHON

ITU and NATO phonetic alphabet

	WORD	SPOKEN		WORD	SPOKEN
A	Alfa	AL fah	N	November	no VEM ber
B	Bravo	BRAH voh	O	Oscar	OSS cah
C	Charlie	CHAR lee	P	Papa	pah PAH
D	Delta	DELL tah	Q	Quebec	keh BECK
E	Echo	ECK oh	R	Romeo	ROW me oh
F	Foxtrot	FOKS trot	S	Sierra	see AIR rah
G	Golf	GOLF	T	Tango	TANG go
H	Hotel	hoh TELL	U	Uniform	YOU nee form
I	India	IN dee ah	V	Victor	VIK tah
J	Juliett	JEW lee ETT	W	Whiskey	WISS key
K	Kilo	KEY loh	X	X-ray	ECKS ray
L	Lima	LEE mah	Y	Yankee	YANG key
M	Mike	MIKE	Z	Zulu	ZOO loo

Numbers

	WORD	SPOKEN		WORD	SPOKEN
0	Zero	ZEE roh	5	Five	FIFE
1	One	WUN	6	Six	SIX
2	Two	TOO	7	Seven	SEV en
3	Three	TREE	8	Eight	AIT
4	Four	FOW er	9	Nine	NIN er

A minimum field exchange, phone

Everything a hunter needs, nothing they do not. Repeat your own callsign at the end of every transmission when you are running.

YOU CQ Parks on the Air, CQ POTA, this is W1ABC, Whiskey One Alfa Bravo Charlie, Kilo dash one two three four, over.

THEM W1ABC, this is K9XYZ, Kilo Nine X-ray Yankee Zulu, you are five nine.

YOU K9XYZ, thanks, you are five seven in Delaware Water Gap. QSL and 73, W1ABC, QRZ.

A minimum field exchange, CW

YOU CQ POTA CQ POTA DE W1ABC W1ABC K

THEM W1ABC DE K9XYZ 599 599 BK

YOU K9XYZ TU 579 579 W1ABC K

THEM TU 73 EE

Running a pileup

- Identify on every transmission. A station tuning in mid-run cannot log an unidentified signal.
- Work partial calls: naming the fragment heard resolves a pileup faster than repeated general calls.
- Read the callsign back while writing it, rather than pausing.
- Announce band changes before making them, so the pileup follows rather than disperses.
- Log times in UTC as contacts are made. Reconstructed timestamps are a common cause of rejected submissions.

05 AWARDS

Activation and Award Requirements

Minimum requirements for the programs a portable station is most likely to be working. The figures below are the ones that determine whether an outing counts.

PROGRAM	MINIMUM FOR CREDIT	THE RULES THAT ACTUALLY CATCH PEOPLE
Parks on the Air (POTA)	10 QSOs from inside the park, in one UTC day	All equipment and operator inside park boundaries, on public land. Within 30.5 m of the feature for trail and river references. Activator uploads an ADIF log. Hunters submit nothing.
Summits on the Air (SOTA)	4 QSOs from inside the activation zone for summit points	Activation zone is within 25 vertical meters of the summit. Station must be wholly independent of any motor vehicle for both power and antenna support. Human powered access. Winter bonus points apply in some associations.
World Wide Flora and Fauna (WWFF)	44 QSOs from inside the reference area	Higher bar than POTA. Many sites hold both a POTA and a WWFF reference, so one log can serve both if you keep working past ten.
Worked All States (WAS)	Confirmed contact with all 50 states	Confirmations via LoTW, physical QSL, or an accepted equivalent. Endorsements exist per band and per mode. A portable station chasing WAS should track by band from the start.
DXCC	100 confirmed entities	Entities are not countries. Check the current DXCC list rather than a map. QRP and portable endorsements exist.
Worked All Continents (WAC)	6 continents confirmed	The most achievable serious HF award from a wire in a park. Africa is usually the last one.

AWARDS

> Contact counts are assessed against a single UTC day, not against a single outing. An activation beginning at 7 p.m. Eastern in summer has already crossed the boundary: eight contacts before 0000Z and eight after produce two incomplete activations rather than one complete one. See Time and the Zulu Day.

Confirm before leaving the site

- Contact count meets the program minimum, counted within a single UTC day, not within your afternoon.
- Every logged time is UTC, and your radio or phone clock was actually correct when you wrote them.
- Park or summit reference number is written down exactly, including the prefix. K-1234 and US-1234 are not interchangeable.
- Your own callsign as used on the air matches what you will submit. Portable indicators matter to some programs and not to others.
- Any park-to-park or summit-to-summit contacts are flagged in the Notes column so you can find them again.

06 TIME

Time and the Zulu Day

Amateur radio is keyed to Coordinated Universal Time. A net at 0100Z meets at the same instant everywhere; one operator's Tuesday evening is another's Wednesday morning, and both log the same time.

North American conversion, UTC to local clock

An asterisk indicates that the local date is one day earlier than the UTC date. This is where activation counts are most often lost.

UTC	EDT (-4)	CDT (-5)	MDT (-6)	PDT (-7)	AKDT (-8)	HST (-10)
00	20	19	18	17	16	15
01	21	20	19	18	17	16
02	22	21	20	19	18	17
03	23	22	21	20	19	18
04	00*	23	22	21	20	19
05	01*	00*	23	22	21	20
06	02*	01*	00*	23	22	21
12	08	07	06	05	04	03
18	14	13	12	11	10	09
21	17	16	15	14	13	12
22	18	17	16	15	14	13
23	19	18	17	16	15	14

Standard time in winter shifts each column one hour further back: EST is UTC minus 5, CST is UTC minus 6, and so on. Hawaii does not observe daylight saving time and stays at UTC minus 10 year round. Arizona, excepting the Navajo Nation, stays at UTC minus 7 year round.

Logging practice

- Set the transceiver clock to UTC and leave it. Conversion in the field is a source of error.
- Enter the UTC date at the head of every sheet before the first contact, and rule a heavy line across the grid at the rollover with the new date beside it.

Spoken form

Say 0100Z as "zero one hundred Zulu" and 1430Z as "fourteen thirty Zulu." On CW, times are usually sent as four digits with no punctuation.

> This copy is keyed to UTC. In summer the UTC day rolls over at 0000 local time in your location, which is the middle of an evening activation. Write the new UTC date on the sheet at that moment or the contacts either side of it will be filed against the wrong day.

TIME

07 GRID

Maidenhead Grid Squares

The Maidenhead locator is a compressed coordinate. Six characters resolve to a few miles, which is sufficient for a VHF contact and short enough to send on CW.

How the locator is built

LEVEL	CHARACTERS	SIZE	EXAMPLE
Field	2 letters, A to R	20 deg longitude by 10 deg latitude	`FN, EM, DM`
Square	2 digits, 0 to 9	2 deg longitude by 1 deg latitude	`FN31, EM73`
Subsquare	2 letters, a to x	5 min longitude by 2.5 min latitude	`FN31pr`
Extended	2 digits	30 sec longitude by 15 sec latitude	`FN31pr55`

Deriving a locator

You will not usually need to. Any handheld GPS, phone app, or modern radio will report it. But the arithmetic is worth understanding once, because it tells you what the letters mean.

1. Take longitude, add 180 so the numbering starts at the antimeridian. Divide by 20. The whole part indexes the first field letter, A through R.
2. Take latitude, add 90. Divide by 10. The whole part indexes the second field letter.
3. Divide the longitude remainder by 2 for the first digit, and take the whole latitude remainder for the second digit.
4. Divide what is left of longitude by 5 minutes for the first subsquare letter and what is left of latitude by 2.5 minutes for the second, a through x.

GRID

> Longitude is always given first. FN31 and 31FN are not the same place, and the second character of the field carries latitude. A locator that resolves to open ocean usually indicates a transposed pair.

Estimating distance

At mid latitudes in North America, one full grid square is roughly 100 miles north to south and roughly 100 miles east to west. Counting squares on a map is accurate enough for deciding whether a 2 meter SSB contact was worth writing home about.

Sites and locators

SITE NAME	GRID (6 CHAR)	COORDINATES

08 PROP

Propagation for the Field Operator

Practical propagation planning reduces to two decisions: which band to work, and whether a given site is worth operating from.

Indices

INDEX	RANGE	WHAT IT TELLS YOU
SFI, solar flux index	65 to 300	Higher is better for the high bands. Below 75 the bands above 20 meters are usually dead. Above 150, 10 and 15 meters open widely.
A index	0 to 400, daily	Geomagnetic disturbance averaged over a day. Below 10 is quiet. Above 20 expect degraded polar and high latitude paths.
K index	0 to 9, every 3 hours	The number to watch. 0 to 2 is quiet, 3 is unsettled, 4 is active, 5 and above is a storm and your high band DX is finished for a while.
MUF	Varies by path and hour	Maximum usable frequency for a given path. Work just below it for the strongest signals.
LUF	Varies by path and hour	Lowest usable frequency. Absorption below it eats you alive, which is why 80 meters is useless at noon.

Band by band, by time of day

A rough guide for a mid latitude North American station at moderate solar activity. Local means under 100 miles, regional means roughly 100 to 1000 miles, DX means beyond that.

BAND	DAWN	MORNING	MIDDAY	AFTERNOON	DUSK	NIGHT	NOTES
160 m	Dead	Dead	Opening	Best	Best	Closing	Winter nights only. Very quiet sites required.
80/75 m	Local	Local	Regional	Regional to DX	DX	Regional	Reliable 300 mile band after dark.
60 m	Local	Local	Regional	Regional	Regional	Regional	Excellent NVIS. Very stable.
40 m	Regional	Regional	Regional to DX	DX	DX	Regional	The one band that works nearly all day.
30 m	Regional	Regional	DX	DX	DX	Regional	CW and data only. Quiet and underused.
20 m	Opening	Best	Best	Best	Closing	Dead	The workhorse for daytime portable operating.
17 m	Opening	Good	Good	Good	Closing	Dead	Less crowded than 20. Try it first.
15 m	Opening	Good	Best	Good	Closing	Dead	Needs SFI above about 110 to be useful.
12 m	Marginal	Good	Good	Marginal	Dead	Dead	Feast or famine. Check it when 15 is open.
10 m	Marginal	Good	Best	Marginal	Dead	Dead	Sporadic E in May to August regardless of the sun.
6 m	Sporadic E	Sporadic E	Sporadic E	Sporadic E	Dead	Dead	June and July. Watch 50.125 constantly.

Grayline

For roughly 30 to 60 minutes at your local sunrise and sunset, the band of twilight sweeping the earth carries low band signals far better than either full day or full night. The D layer, which absorbs, has decayed; the F layer, which refracts, has not. Paths along that line open dramatically. If you are chasing distance on 40 or 80 meters from a park, arrive early enough to catch your own sunset.

Near vertical incidence skywave

Near vertical incidence skywave fills the gap between ground wave and the first skip zone, roughly 0 to 300 miles. Put a dipole low, a tenth of a wavelength up or less, on 80, 60, or 40 meters, and the pattern fires almost straight up. It comes back down over a wide circle with no dead zone. This is the correct configuration for regional emergency traffic and for working your own state, and it is the one time in amateur radio when a low antenna is the right antenna.

Site noise is a propagation factor. A site three decibels quieter than the usual noise floor yields more than doubling transmitter power. Record the measured noise floor on every Site Record; sites reading S1 on 40 meters are worth the travel.

09 ANT

Antenna Math for the Field

Wire is the portable station's principal advantage: light, cheap, and, at a half wavelength in a tree, consistently better than a shortened whip at vehicle height.

Wire lengths, in feet

Half wave uses 468 divided by frequency in megahertz. Quarter wave uses 234. Full wave, for end fed half wave transformers used on their harmonic bands and for loops, uses 936. These are rules of thumb that already include an end effect correction. Cut long by three percent, measure, then trim.

BAND	DESIGN F (MHZ)	HALF WAVE (FT)	QUARTER WAVE (FT)	FULL WAVE (FT)	FIELD NOTES
160	1.900	246.3	123.2	492.6	Full size is rarely portable. Consider a loaded vertical.
80	3.650	128.2	64.1	256.4	A sloped or inverted V fits most parks.
60	5.360	87.3	43.7	174.6	Excellent NVIS at 15 to 25 feet.
40	7.150	65.5	32.7	130.9	The standard portable dipole.
30	10.125	46.2	23.1	92.4	Fits between the trees almost anywhere.
20	14.200	33.0	16.5	65.9	The one to build first.
17	18.118	25.8	12.9	51.7	
15	21.300	22.0	11.0	43.9	
12	24.940	18.8	9.4	37.5	
10	28.400	16.5	8.2	33.0	A full wave EFHW here is only 33 feet.
6	50.150	9.3	4.7	18.7	Short enough to hold up on a painter's pole.

Height and pattern

- A horizontal dipole at a quarter wavelength or lower fires most of its energy up. That is NVIS, and it is the right choice for regional work.
- The same dipole at a half wavelength drops the main lobe to roughly 30 degrees and starts to work DX.
- Above one wavelength, gains are real but small and the rigging cost rises fast. In a park, spend your effort on getting the feedpoint clear of the ground rather than on the last five feet of height.
- A quarter wave vertical needs radials. Four elevated radials cut for the band beat sixteen random wires on the dirt. With no radials at all, your coax shield becomes the counterpoise and the pattern is whatever the coax feels like.
- An end fed half wave with a 49 to 1 transformer still needs a counterpoise, usually a length of wire around 0.05 wavelength, or the feedline does the job badly.

SWR, return loss and mismatch loss

The reflected power figure is the fraction of forward power that comes back from the mismatch. In a low loss line, most of it is re-reflected by the transmitter and eventually radiated, so a 2 to 1 match is not wasting eleven percent of your signal. What high SWR really costs you is transmitter power foldback and extra loss in lossy coax.

SWR	REFLECTION COEFF	REFLECTED POWER	RETURN LOSS	MISMATCH LOSS (DB)
1.0 : 1	0.000	0.0 %	Infinite	0.00
1.2 : 1	0.091	0.8 %	20.8 dB	0.04
1.5 : 1	0.200	4.0 %	14.0 dB	0.18
2.0 : 1	0.333	11.1 %	9.5 dB	0.51
2.5 : 1	0.429	18.4 %	7.4 dB	0.88
3.0 : 1	0.500	25.0 %	6.0 dB	1.25
4.0 : 1	0.600	36.0 %	4.4 dB	1.94
5.0 : 1	0.667	44.4 %	3.5 dB	2.55
10.0 : 1	0.818	66.9 %	1.7 dB	4.81

Feedline, nominal loss per 100 feet, matched

CABLE	VF	Z0	14 MHZ	28 MHZ	50 MHZ	146 MHZ	440 MHZ
RG-58 solid PE	0.66	50	1.8	2.6	3.3	6.2	11.5
RG-8X foam PE	0.82	50	1.2	1.7	2.3	4.0	7.4
RG-213 solid PE	0.66	50	0.6	0.9	1.2	2.2	4.2
LMR-400 type	0.85	50	0.4	0.6	0.8	1.5	2.7
RG-174 thin	0.66	50	5.5	7.5	9.5	17.0	31.0
450 ohm window	0.91	450	0.05	0.08	0.10	0.25	0.55

> Losses accumulate without being obvious. Fifty feet of RG-174 at 146 MHz dissipates more than half the transmitter output before the antenna. For a low power field station a short run of RG-8X is generally the best compromise between weight and loss. Velocity factor applies when cutting a matching stub or a quarter wave transformer: multiply the free space length by VF.

10 POWER

Power Budget, Battery, and Wiring

A field station is limited by its supply more often than by propagation or equipment. Both the capacity and the wiring have to be sized to the actual load.

Typical current draw at 13.8 V

STATE	CURRENT	NOTES
Receive, headphones	0.5 - 1.0 A	Most of a long day is spent here.
Receive, internal speaker	0.8 - 1.5 A	Headphones save real amp-hours.
Transmit, 5 W SSB	1.5 - 2.5 A	Average on voice is far below peak.
Transmit, 10 W SSB	2.5 - 4 A	
Transmit, 100 W SSB	18 - 22 A peak	Voice duty cycle keeps the average near a third of this.
Transmit, 100 W CW	15 - 20 A key down	Key down is continuous. Budget for it.
Transmit, 100 W digital	18 - 22 A continuous	Worst case. Full duty cycle, and it heats the finals.
Tablet or logging device	0.5 - 1.5 A	Charge it from the same battery and it is part of the budget.

Runtime

Estimate an average current, not a peak. For a normal operating session, assume you transmit about one third of the time on phone and about half the time on CW. Then:

```
Average A = (RX amps x RX fraction) + (TX amps x TX fraction)
Runtime h = (Battery Ah x usable fraction) / Average A
```

Usable fraction depends on chemistry. Lithium iron phosphate will give you roughly 90 percent of rated capacity with a flat voltage curve almost to the end. Sealed lead acid should be treated as 50 percent usable if you want the battery to survive the year, and its voltage sags under transmit load in a way that will make a radio unhappy long before the amp-hours run out.

Runtime worksheet

CONFIGURATION	AVG CURRENT (A)	BATTERY (AH USABLE)	PREDICTED HOURS

POWER

Wire gauge and fusing for 12 volt DC

Fuse to protect the wire, not the radio. The fuse should open before the smallest conductor in the branch reaches its rating. Fuse the positive lead as close to the battery terminal as physically possible: everything downstream of that point is unprotected.

GAUGE	PRACTICAL DC LIMIT	OHMS PER FOOT	TYPICAL USE IN A FIELD STATION
18 AWG	10 A	0.0064	Accessory leads, control lines
16 AWG	13 A	0.0040	Light accessory runs
14 AWG	17 A	0.0025	QRP main feed, short runs
12 AWG	23 A	0.0016	100 W radio main feed, under 6 feet
10 AWG	33 A	0.0010	100 W radio, longer runs, battery to panel
8 AWG	46 A	0.00063	Battery main and disconnect

Lithium iron phosphate

- Do not charge below freezing. LiFePO4 will accept charge at low temperature and plate lithium doing it, which permanently destroys capacity. Discharging cold is fine.
- The internal battery management system is a protection device, not a charge controller and not a substitute for a fuse. When it trips, it disconnects the whole pack, usually in the middle of your best contact.
- A LiFePO4 pack holds near 13.2 volts across most of its discharge and then falls off a cliff. Voltage is a poor fuel gauge. Track amp-hours or carry a coulomb counter.
- Charge to storage voltage, not full, if the pack will sit for months.
- Carry the pack in a hard case with the terminals covered. A wrench across a 20 amp-hour pack will weld itself to the terminals.

11 SAFE

RF Exposure and Field Safety

Since 3 May 2023 every amateur station in the United States must have a current RF exposure evaluation, at any power level. The categorical exemption by power was removed. The evaluation is retained by the licensee and is not filed with anyone.

Maximum permissible exposure, mW per square centimeter

Controlled means you and people who know the transmitter is there and can act accordingly. Uncontrolled means everybody else, which in a public park means everybody else.

FREQUENCY	CONTROLLED (MPE)	UNCONTROLLED (MPE)	NOTES
0.3 - 1.34 MHz	100	180 / f squared	f in MHz
1.34 - 3.0 MHz	900 / f squared	180 / f squared	
3.0 - 30 MHz	900 / f squared	180 / f squared	At 14 MHz: 4.59 controlled, 0.92 uncontrolled
30 - 300 MHz	1.0	0.2	The most restrictive range
300 - 1500 MHz	f / 300	f / 1500	
1500 - 100,000 MHz	5.0	1.0	

Evaluating a portable station

1. Record your maximum power at the antenna feedpoint, your antenna gain, your duty cycle, and the lowest frequency you will use.
2. Compute or look up the minimum safe distance for uncontrolled exposure at that power and frequency.
3. Set up so that the nearest point the public can reach exceeds that distance. In practice this means the feedpoint and the high current portion of the wire are out of arm's reach of a trail.
4. Write the result down. The Site Record page in Part Two has a field for the exclusion distance you used.

> In a public area the governing hazard is contact with the high voltage end of an end fed wire, not diffuse field exposure at the operating position. Keep wire ends above head height and clear of any path.

Weather and lightning

- You are erecting a conductor on a mast in an open field. Treat the first distant thunder as the signal to take it down, not as a prompt to check the radar.
- Thirty minutes after the last thunder before you put anything back up.
- A fiberglass mast is not an insulator once it is wet.
- Disconnect the feedline from the radio when you break for lunch. It costs nothing.

Site and personal safety

- Look up before you throw a line. Power lines end more amateur careers than anything else in this book. If a wire could reach a power line when it falls, the site is wrong.
- Mark guy lines and radials with flagging tape. In a public park, somebody will walk into them.

- Water, shade, and a hat outlast enthusiasm. Cold weather operating fails through hands, not through gear.
- Leave no trace. Pack out the wire ties, the electrical tape, and the little plastic caps off the connectors. Amateur radio's access to public land is a courtesy, and it is revocable.
- Tell somebody where you are going and when you will be back, particularly for summits. A handheld on 146.520 is not a rescue plan.

12 SECT

ARRL and RAC Sections

The section abbreviation forms the exchange in Field Day, Sweepstakes and the Simulated Emergency Test. Sections are not states: several states are divided, and some sections cross state lines.

CALL AREA 0	
CO	Colorado
IA	Iowa
KS	Kansas
MN	Minnesota
MO	Missouri
ND	North Dakota
NE	Nebraska
SD	South Dakota
CALL AREA 1	
CT	Connecticut
EMA	Eastern Massachusetts
ME	Maine
NH	New Hampshire
RI	Rhode Island
VT	Vermont
WMA	Western Massachusetts
CALL AREA 2	
ENY	Eastern New York
NLI	New York City-Long Island
NNJ	Northern New Jersey
NNY	Northern New York
SNJ	Southern New Jersey
WNY	Western New York
CALL AREA 3	
DE	Delaware
EPA	Eastern Pennsylvania
MDC	Maryland-DC
WPA	Western Pennsylvania
CALL AREA 4	
AL	Alabama
GA	Georgia
KY	Kentucky
NC	North Carolina
NFL	Northern Florida
PR	Puerto Rico
SC	South Carolina
SFL	Southern Florida
TN	Tennessee
VA	Virginia
VI	US Virgin Islands
WCF	West Central Florida
CALL AREA 5	
AR	Arkansas
LA	Louisiana
MS	Mississippi
NM	New Mexico
NTX	North Texas
OK	Oklahoma
STX	South Texas
WTX	West Texas
CALL AREA 6	
EB	East Bay
LAX	Los Angeles
ORG	Orange
PAC	Pacific
SB	Santa Barbara
SCV	Santa Clara Valley
SDG	San Diego
SF	San Francisco
SJV	San Joaquin Valley
SV	Sacramento Valley
CALL AREA 7	
AK	Alaska
AZ	Arizona
EWA	Eastern Washington
ID	Idaho
MT	Montana
NV	Nevada
OR	Oregon
UT	Utah
WWA	Western Washington
WY	Wyoming
CALL AREA 8	
MI	Michigan
OH	Ohio
WV	West Virginia
CALL AREA 9	
IL	Illinois
IN	Indiana
WI	Wisconsin
CANADA (RAC)	
AB	Alberta
BC	British Columbia
GH	Ontario Golden Horseshoe
MB	Manitoba
NB	New Brunswick
NL	Newfoundland and Labrador
NS	Nova Scotia
ONE	Ontario East
ONN	Ontario North
ONS	Ontario South
PE	Prince Edward Island
QC	Quebec
SK	Saskatchewan
TER	Territories

Florida holds three sections: Northern, Southern and West Central. Massachusetts, New York, New Jersey, Pennsylvania, Texas, Washington and California are also divided. Establish which section a site falls in before operating near a boundary.

13 PFX

Common Callsign Prefixes

A working subset, sufficient to place a signal before checking it properly. DXCC counts entities rather than countries, and the official list is longer than this page.

US call areas

The numeral in a US callsign originally indicated the licensee's call area, and for vanity and older calls it usually still reflects where the licensee lived when the call was issued. It has not been a reliable location indicator for decades. Ask for a state.

0	1	2	3	4	5	6	7	8	9
CO IA KS MN MO ND NE SD	CT MA ME NH RI VT	NY NJ	DE PA MD DC	AL FL GA KY NC SC TN VA	AR LA MS NM OK TX	CA	AK AZ ID MT NV OR UT WA WY	MI OH WV	IL IN WI

By region and entity

REGION	PREFIX	ENTITY	PREFIX	ENTITY
North America	`K, N, W, AA-AL`	United States	`VE, VA, VO, VY`	Canada
	`XE, XF`	Mexico	`KP4, NP4, WP4`	Puerto Rico
	`KH6, NH6, WH6`	Hawaii	`KL7, AL7, NL7`	Alaska
	`CO, CM`	Cuba	`HI`	Dominican Republic
South America	`PY, PP-PW`	Brazil	`LU`	Argentina
	`CE`	Chile	`HK`	Colombia
	`YV`	Venezuela	`OA`	Peru
Europe	`G, M, 2E`	England	`GM, MM`	Scotland
	`EI`	Ireland	`F`	France
	`DL, DA-DR`	Germany	`I`	Italy
	`EA`	Spain	`CT`	Portugal
	`PA`	Netherlands	`ON`	Belgium
	`SM`	Sweden	`LA`	Norway
	`OH`	Finland	`OZ`	Denmark
	`SP`	Poland	`OK`	Czech Republic
	`HA`	Hungary	`YO`	Romania
	`UA, R`	Russia	`UR`	Ukraine
	`S5`	Slovenia	`9A`	Croatia
	`HB9`	Switzerland	`OE`	Austria
Africa	`ZS`	South Africa	`CN`	Morocco
	`SU`	Egypt	`5N`	Nigeria
	`5Z`	Kenya	`EA8`	Canary Islands
Asia	`JA-JS`	Japan	`BY, BA-BL`	China
	`HL, DS`	South Korea	`VU`	India
	`9V`	Singapore	`HS`	Thailand
	`4X, 4Z`	Israel	`TA`	Turkey
Oceania	`VK`	Australia	`ZL`	New Zealand
	`KH2`	Guam	`FK`	New Caledonia
	`YB`	Indonesia	`DU`	Philippines

14 CHECK

Pre-Deployment and Teardown Checklists

Read aloud and confirmed item by item. A checklist that is skimmed records only what the operator already assumes.

Pre-deployment

License and identity

- ☐ Callsign known and correct for the mode of operation
- ☐ Any portable indicator decided before you leave
- ☐ Award program reference number written down, with prefix

Radio and control

- ☐ Radio, microphone, key or paddle, and their cables
- ☐ Clock set to UTC and verified
- ☐ Power output set to the band and program limits
- ☐ Spare fuses for every fused device

Antenna

- ☐ Wire, transformer or balun, and insulators
- ☐ Throw line, weight, and a second throw line
- ☐ Mast, guys, stakes, and something to drive them with
- ☐ Coax and the adapters you actually need, not all of them
- ☐ Antenna analyzer or SWR meter

Power

- ☐ Battery charged, and the charge state actually verified
- ☐ Fused leads, Powerpole or equivalent, no bare terminals
- ☐ Second battery or a charging plan for a long day

Logging

- ☐ This book, and two pencils
- ☐ Electronic log ready and its device charged
- ☐ Paper backup of the reference number and any schedule

Personal and site

- ☐ Water, food, sun and insect protection, layers
- ☐ Site permission and hours confirmed
- ☐ Somebody knows where you are and when you will be back
- ☐ Phone charged, and you know whether it will have coverage

Teardown and follow-up

Before closing down

- ☐ Contact count meets the program minimum inside one UTC day

- ☐ Every time in the log is UTC
- ☐ Park to park and summit to summit contacts flagged
- ☐ Debrief page started while the details are fresh

Teardown

- ☐ Antenna down, wire coiled, nothing left in a tree
- ☐ All stakes, guys, and flagging recovered
- ☐ Battery disconnected and terminals covered
- ☐ Connectors capped, coax coiled without kinks
- ☐ Site walked once with a light, looking for what you dropped

Within 24 hours

- ☐ Log uploaded to the program and to LoTW
- ☐ Anything that failed written into the Modification and Repair log
- ☐ Anything that needs replacing added to the go-kit list
- ☐ Battery on charge

COLOPHON

The interior is set in IBM Plex: Plex Serif for reading matter, Plex Sans and Plex Sans Condensed for labels and rules, Plex Mono for tabular matter. Log grids are ruled at 0.4 point, a weight that carries pencil, ballpoint and fine-line ink without competing with the entry. Row heights are set for handwriting done in the field rather than at a desk.

Section markers are printed rather than die cut, so the book can be produced by any printer without specialist finishing.

Sources for Part One

- 47 CFR Part 97, Amateur Radio Service, US Federal Communications Commission.
- ARRL frequency allocations and band plan charts, and ARRL/RAC contest section list.
- FCC Report and Order on 60 meters, new 5351.5 to 5366.5 kHz subband effective 13 February 2026.
- FCC Report and Order eliminating HF symbol rate limits in favor of a 2.8 kHz bandwidth limit, effective 2024.
- Parks on the Air program rules and activator documentation.
- Summits on the Air general rules.
- FCC Office of Engineering and Technology Bulletin 65 on RF exposure, and the amateur exposure evaluation requirement in effect since 3 May 2023.

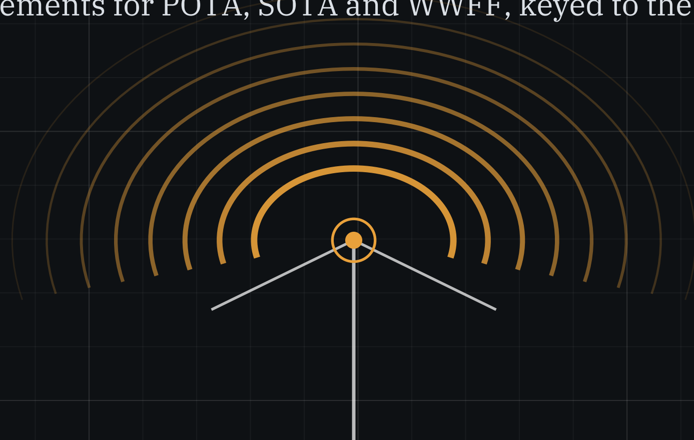
Coax loss, velocity factor, and conductor ampacity figures are nominal values for typical cable and wire of each type at room temperature. Use them for planning. Measure for anything that matters.

A logbook for a station that is carried in, assembled on site, and taken down the same day.

FIELD RECORD is organized around the outing rather than the contact. Twenty-six deployments, four pages each: a Site Record completed before the first call, two sheets of contacts in the columns portable operation requires, and a debrief written at the site. Part One is a field reference compiled from primary sources for operators working away from mains power. Part Three holds the station records that carry forward from one volume to the next.

IN PART ONE

- US band plan and privileges in color, including the 60 meter allocation as revised in February 2026
- South Carolina repeaters and nets, and a quick reference for the Icom IC-706MKII
- Antenna field math, coax loss, SWR tables, and power budget worksheets
- RF exposure limits and the station evaluation required of every licensee
- Propagation indices, grid squares, Q signals, prosigns, phonetics, and ARRL sections
- Activation requirements for POTA, SOTA and WWFF, keyed to the UTC day



SAMPLE "NOCALL" STATION

· FIRST EDITION · 26 DEPLOYMENTS · 40 PAGES